

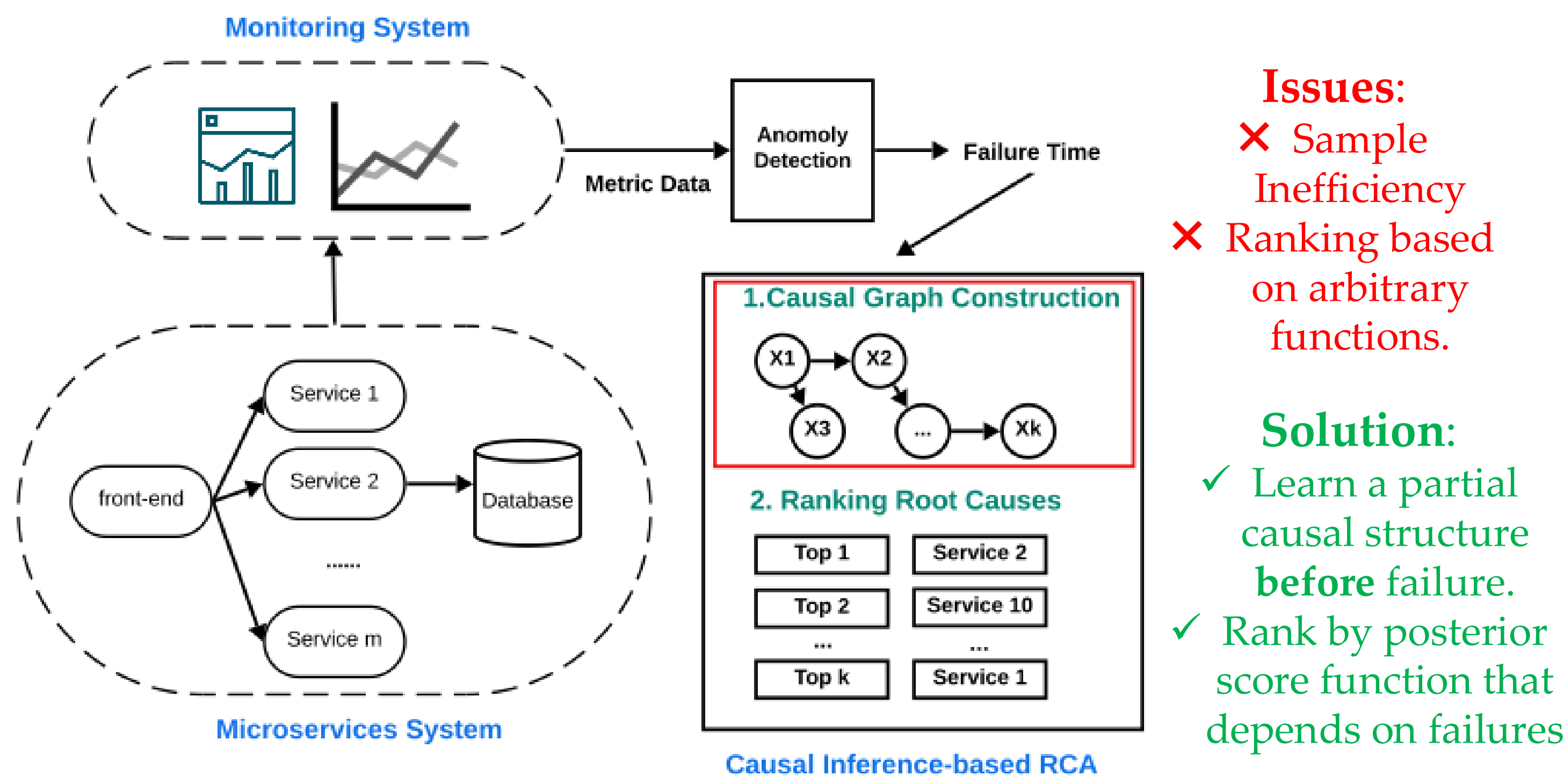
Root Cause Analysis of Failures in Microservices via Bayesian Root Cause Discovery

Kenneth Lee, Zihan Zhou, Murat Kocaoglu

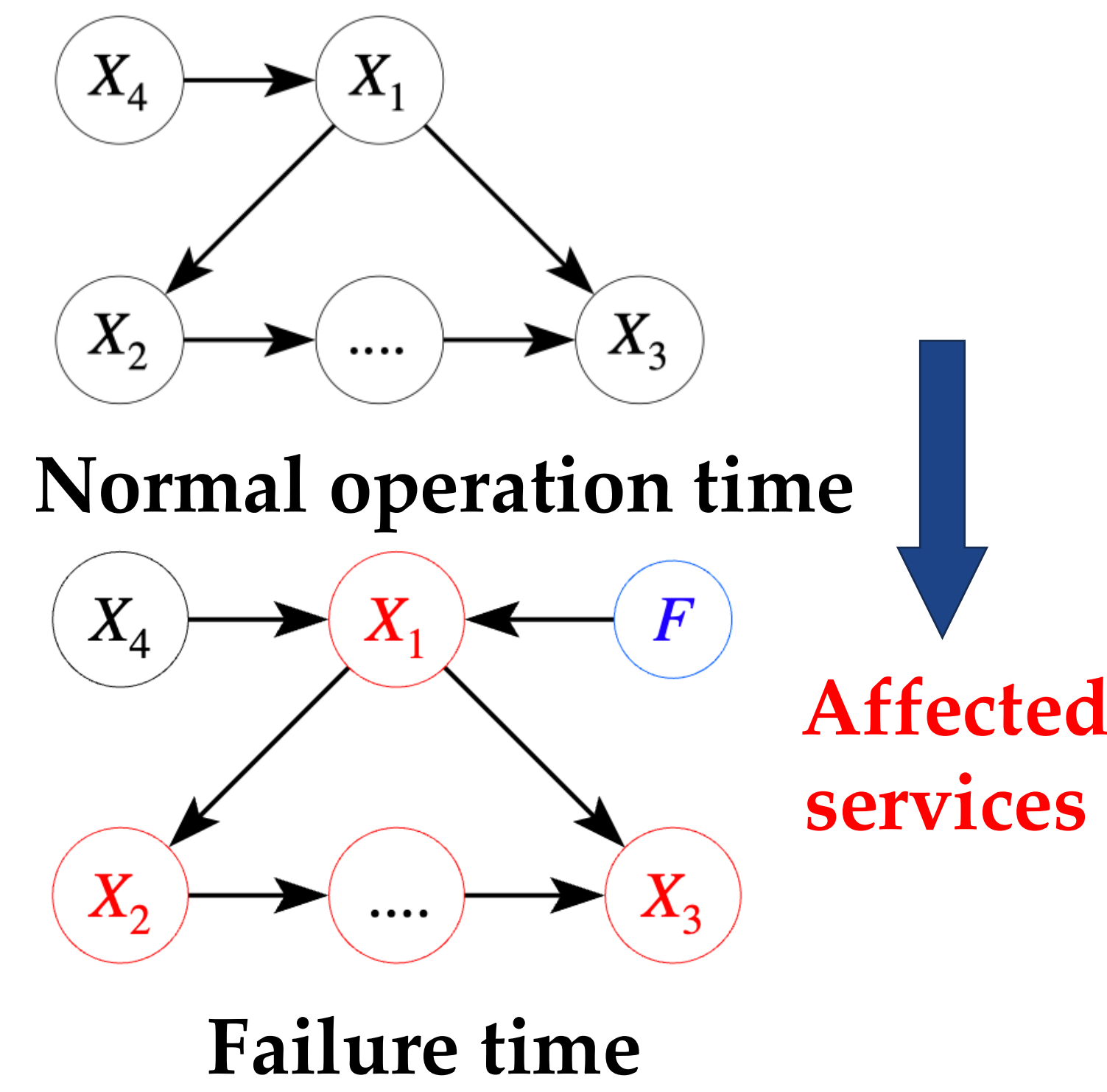
Purdue University, Johns Hopkins University



What is Root Cause Analysis?



Modeling Failures as Interventions



Bayesian Inference of the Root Causes

$$P(R | Data) = \sum_C P(R | C, Data) P(C | Data)$$

$$P(R | C, Data) = \frac{P(Data | R, C) P(R | C)}{\sum_{R'} P(Data | R', C) P(R' | C)}$$

$$P(Data | R, C) = \sum_{G \in C} P(Data | G, R) P(G | C, R)$$

Evaluate the likelihood by sampling a DAG from C

Uniform over all sampled CPDAGs

$$P(C | Data) \propto \frac{P(Data | C) P(C)}{q(C)}$$

Empirical probability from bootstrapping

Robustness to finite sample approximation

Lemma (identifiability). Under modularity, positivity, for almost all parameter values, any two distinct sets $R \neq R'$, if (G^*, R^*) is the ground truth, then

$$\Delta_{\min} := \inf_{(G, R) \neq (G^*, R^*)} \mathbb{E}_{p^*} \left[\log \frac{p(\mathbf{X} | G^*, R^*)}{p(\mathbf{X} | G, R)} \right] > 0.$$

Theorem (Posterior Consistency). Let R^* be a set of root causes. Under A1-A2 and causal sufficiency, we have

$$p(G^*, R^* | \mathcal{D}) \xrightarrow[n \rightarrow \infty]{p^*} 1.$$

Theorem (Finite Sample Bound with ε robustness). For any $\delta \in (0, 1)$, let M be the number of wrong pairs (G, R) and let

$$\Delta_{\min}^{eff}(n) := \Delta_{\min} - 2B\varepsilon, t_n := B\sqrt{\frac{2\ln(2M/\delta)}{n}}$$

, with probability at least $1 - \delta - \eta$,

$$p(G^*, R^* | \mathcal{D}) \geq 1 - M \exp \left\{ -n(\Delta_{\min}^{eff}(n) - t_n) \right\} \max_{(G, R) \neq (G^*, R^*)} \frac{p(G, R)}{p(G^*, R^*)}.$$

Assumptions

- A1 (ε -close plug in): With probability at least $1 - \eta$, $\eta \in (0, 1)$, n is the sample size, $d_{TV}(p', p^*) \leq \varepsilon$, where $\varepsilon \rightarrow 0$ as $n \rightarrow \infty$

- A2 (Bounded log-likelihood ratios). There exists $B \leq \infty$ such that for all (G, R)

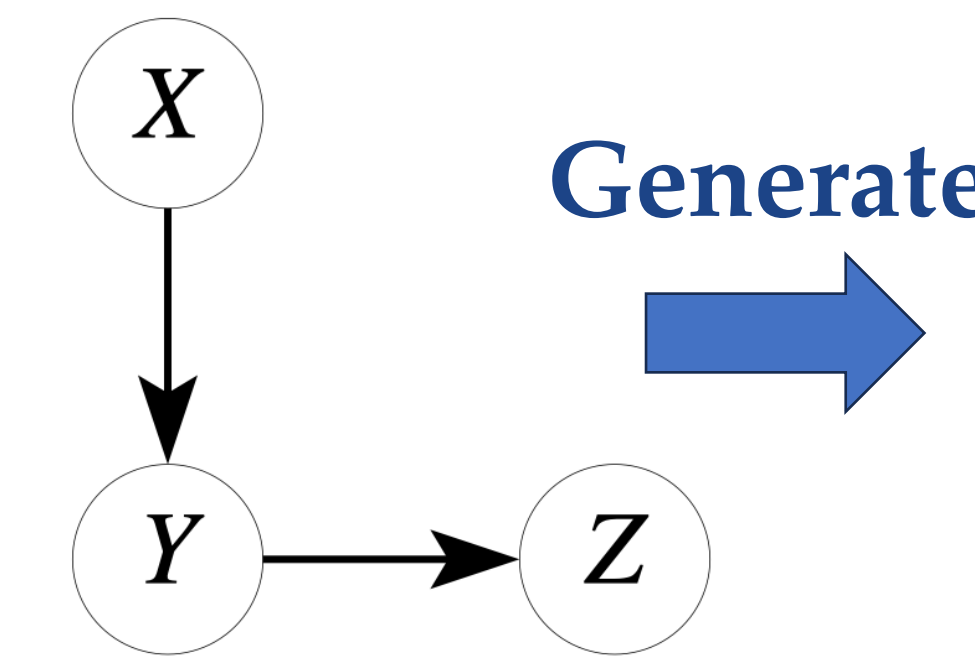
$$\left| \log \frac{p(\mathbf{X} | G^*, R^*)}{p(\mathbf{X} | G, R)} \right| \leq B.$$

Corollary. For any candidate target set R , BRCD enumerates all and only the I-CPDAG compatible with $(C(G^*), R)$ by ranging over all possible orientations of the edge cut $E[R, V \setminus R]$ and closing under Meek rules. Hence, BRCD covers all I-MEC consistent with $(C(G^*), R)$

Lemma. Let G be a DAG and R be a set of root cause candidates. Consider (G, R) as a joint event. With a uniform prior on (G, R) , ranking the posteriors of (G, R) is the same as ranking I-CPDAGs.

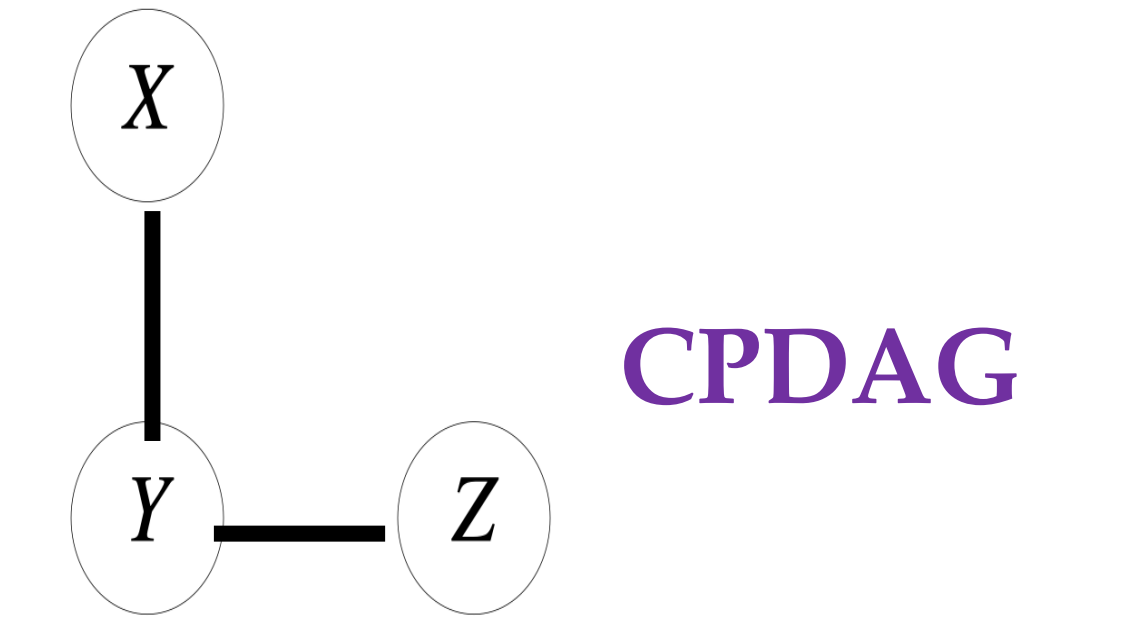
Proposed Algorithm (BRCD)

Before Failure

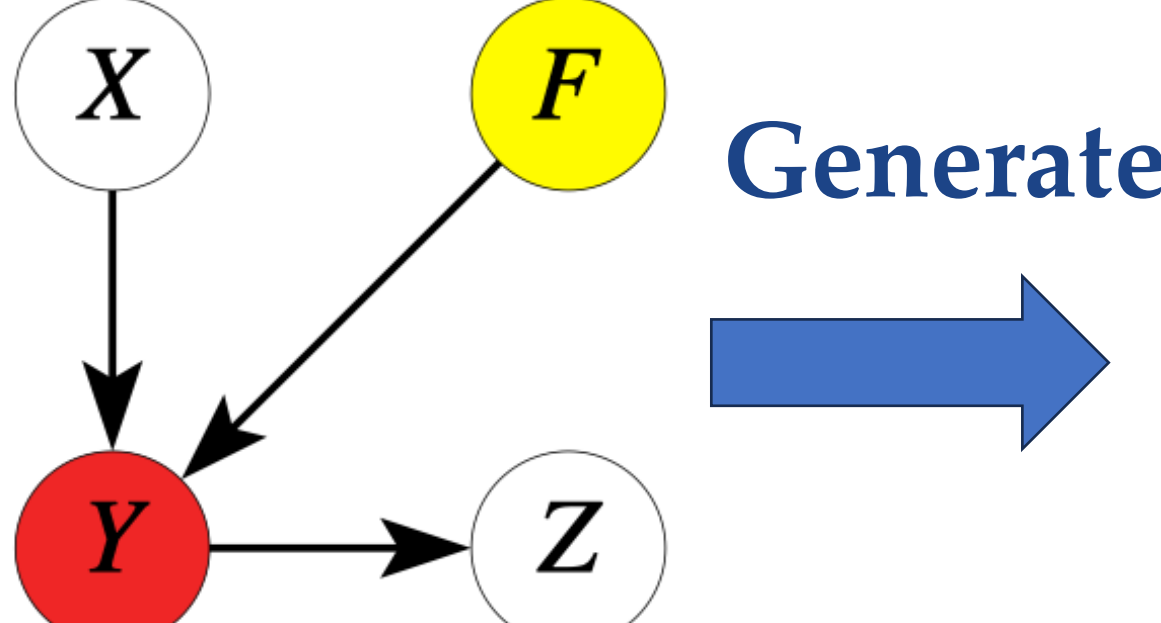


Step 1: Observational Causal Discovery

Bootstrapping (Optional) to compute $P(C | Data)$

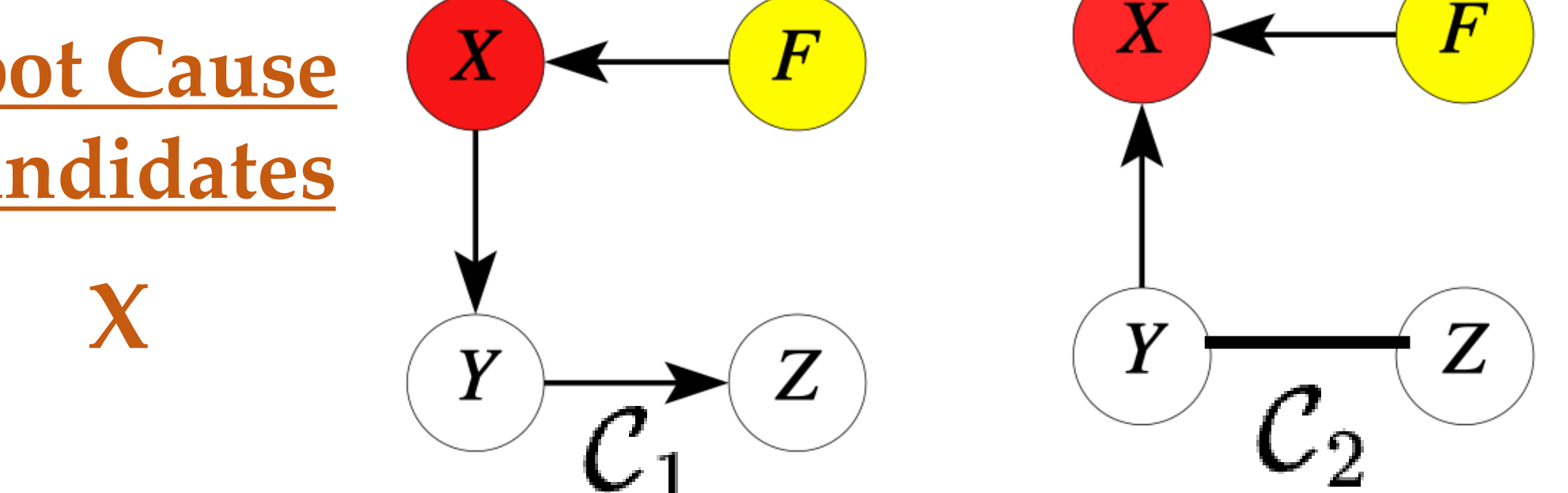


After Failure



Step 2: Enumerate I-CPDAGs $\mathcal{C}_i$

Root Cause Candidates



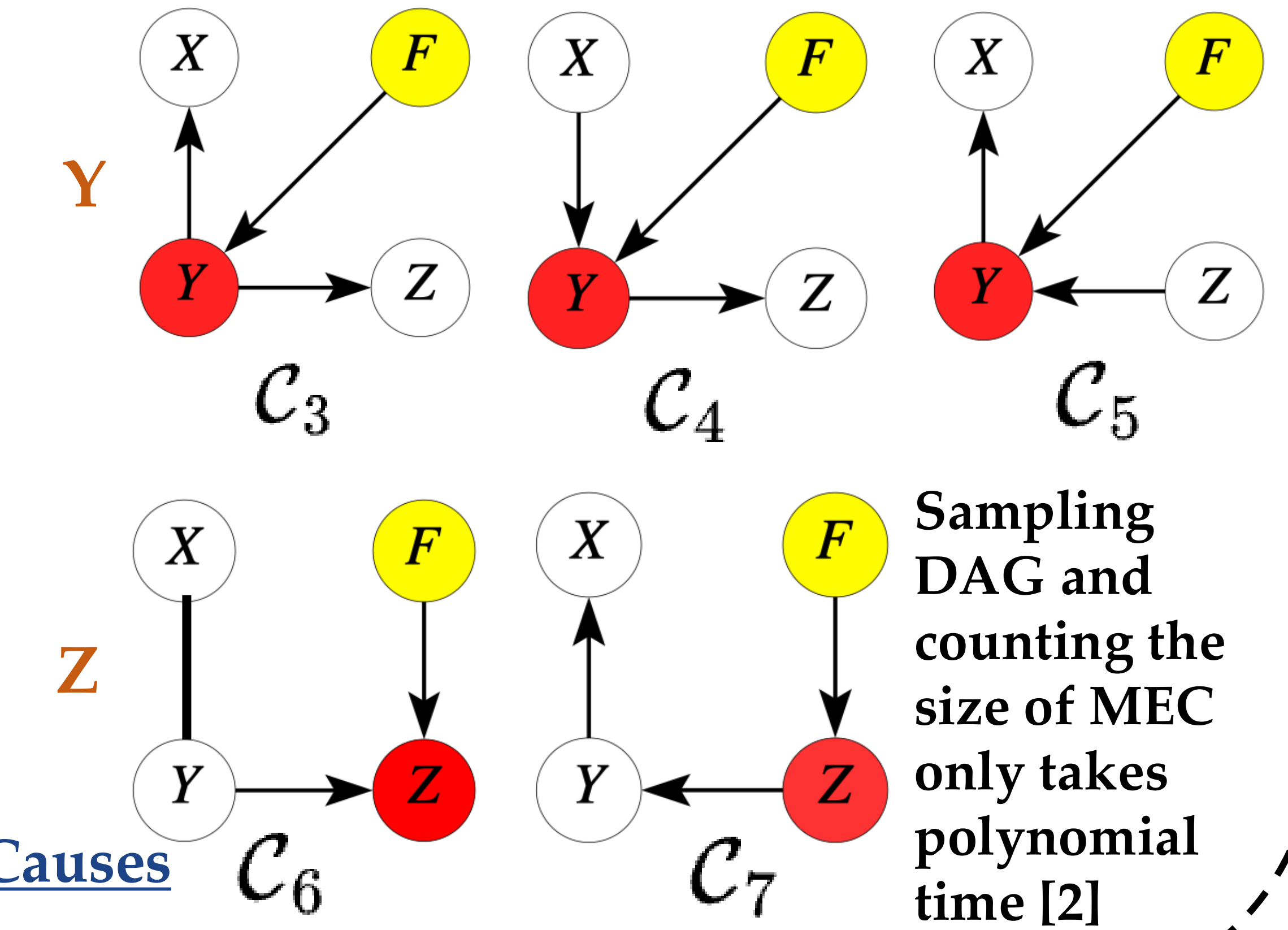
Step 3: Compute Likelihood by Sampling DAGs

For a candidate X

$$P(D | R = X) = P(Z | Y) P(Y | X) P(X | F) P(F) (1/9) + P(Z | Y) P(Y) P(X | Y, F) P(F) (2/9)$$

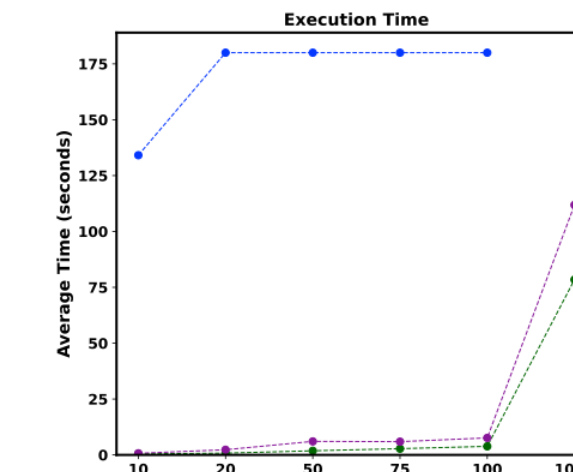
Factorize according the augmented DAGs sampled from $\mathcal{C}_1$ and $\mathcal{C}_2$

Step 4: Rank Root Causes by Posterior $P(R | D)$

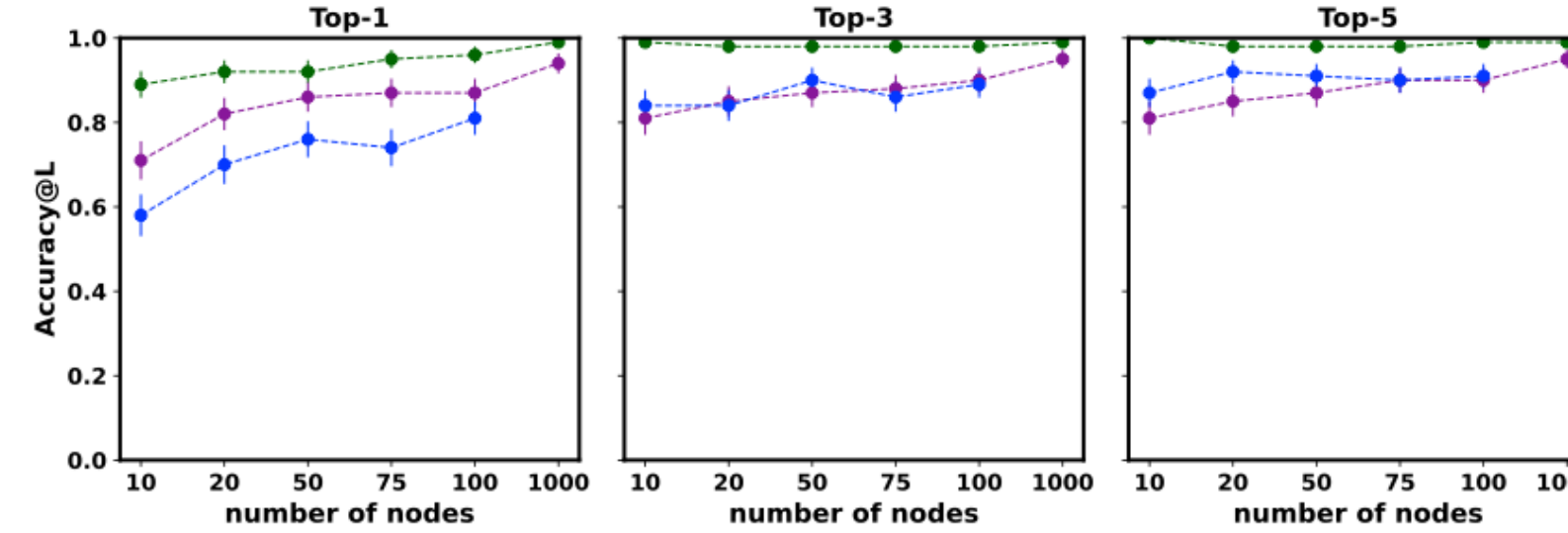


Synthetic Experiment

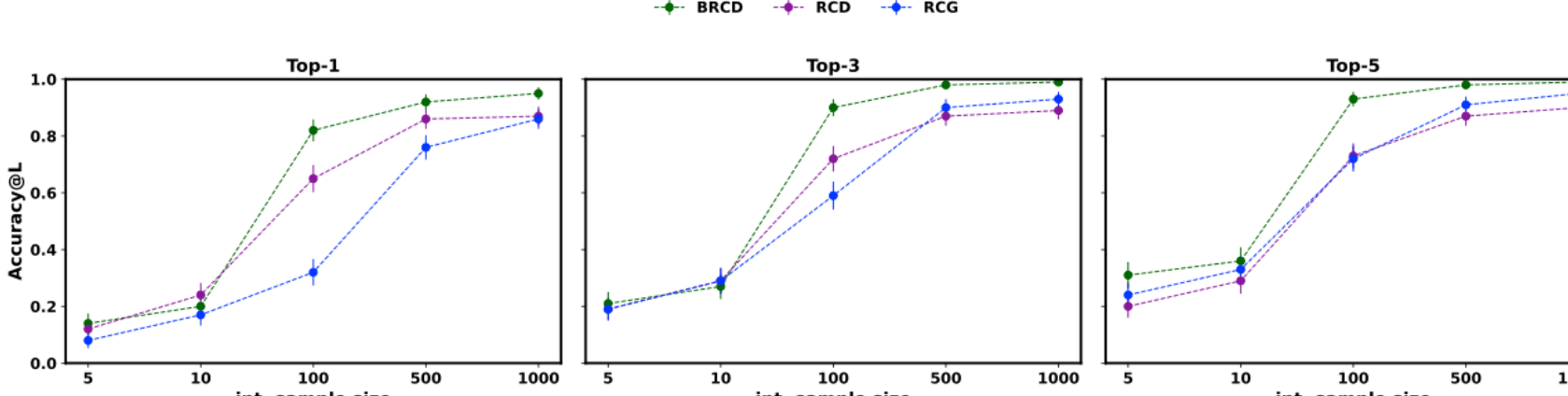
Scalability



Accuracy



Int. Sample Convergence



Experimental Results

Real-world Experiment with Petshop Data

	RCD	RCG	BARO	BRCD	BRCD-C	BRCD-M	BRCD-B10	BRCD-B100	SO	ST	Cholesky	IDI	ShapleyIQ	MicroDig	SimplerCA
high_traffic	0.00	0.00	0.15	0.27	0.35	0.08	0.31	0.31	0.00	0.00	0.00	0.12	0.08	0.00	0.08
low_traffic	0.31	0.15	0.27	0.46	0.42	0.35	0.42	0.46	0.04	0.00	0.00	0.12	0.15	0.00	0.15
temporal_traffic1	0.25	0.12	0.25	0.38	0.63	0.63	0.25	0.50	0.00	0.00	0.00	0.25	0.12	0.00	0.25
temporal_traffic2	0.62	0.25	0.25	0.50	0.88	0.88	0.62	0.62	0.00	0.00	0.00	0.25	0.25	0.00	0.25
Average	0.30	0.13	0.23	0.40	0.57	0.48	0.40	0.47	0.01	0.00	0.00	0.19	0.15	0.00	0.18
high_traffic	0.00	0.00	0.23	0.35	0.38	0.23	0.31	0.35	0.00	0.00	0.00	0.23	0.23	0.42	0.19
low_traffic	0.38	0.19	0.42	0.65	0.69	0.58	0.77	0.73	0.04	0.04	0.00	0.23	0.38	0.62	0.31
temporal_traffic1	0.75	0.25	0.38	0.62	0.75	0.75	0.62	0.75	0.12	0.00	0.00	0.38	0.38	0.62	0.38
temporal_traffic2	0.75	0.25	0.38	0.62	0.88	0.88	0.88	0.88	0.00	0.00	0.00	0.50	0.50	0.62	0.38
Average	0.47	0.17	0.35	0.56	0.68	0.61	0.65	0.68	0.04	0.01	0.00	0.34	0.37	0.57	0.32
high_traffic	0.00	0.23	0.31	0.38	0.42	0.38	0.35	0.42	0.00	0.00	0.00	0.35	0.31	0.62	0.35
low_traffic	0.42	0.31	0.46	0.73	0.73	0.58	0.81	0.85	0.08	0.23	0.04	0.35	0.46	0.65	0.46
temporal_traffic1	0.75	0.38	0.38	0.62	0.75	0.75	0.62	0.75	0.25	0.00	0.00	0.38	0.38	0.62	0.38
temporal_traffic2	0.75	0.38	0.50	0.75	0.88	0.88	0.88	0.88	0.00	0.00	0.12	0.62	0.62	0.62	0.38
Average	0.48	0.33	0.41	0.62	0.69	0.65	0.67	0.73	0.08	0.06	0.04	0.43	0.44	0.63	0.39

Reference

- [1] Ikram, Azam, et al. "Root cause analysis of failures in microservices through causal discovery." Advances in Neural Information Processing Systems 35 (2022): 31158-31170.
- [2] Wienöbst, M., Bannach, M., and Li' skiewicz, M. Polynomial-time algorithms for counting and sampling markov equivalent dags with applications. Journal of Machine Learning Research, 24(213):1-45, 2023.

Paper



Code

